Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project tested: Park It (Parking Management System) (PHP + PostgreSQL) | Date of testing: 7 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	Park It (Parking Management System)
Owning group	Group 2 (member names are not listed in the project files)
Repository	park-it-database (project zip park-it-database-main; the repository link is not included in the files we received)
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a6_backup.txt, a6_restore.txt, a6_verify.txt, a6_access.txt, a6_recovery.txt

Every result below comes from this run on a freshly created database.

Run at about 00:41 on 7 November 2026 (Asia/Manila). The database was in the state described in Activity 4, including the sequence grant added by our group. The original database and the dump were not modified by the restore and recovery tests.

2. Backup plan

Item	Plan
Objects	Whole database grp_parkitdb: 5 tables, 5 sequences, 9 indexes, 3 foreign keys and the grants
Method	pg_dump custom format (-Fc). The project has no backup procedure, so this is the group's own plan
Account	The project defines no backup role, so the dump was taken as postgres. Recommend a role with pg_read_all_data
Files outside the database	config\database.php (connection settings) and the PHP files themselves, which are in Git
Schedule (real deployment)	Daily full dump at closing time, plus one before any change
Retention	7 daily and 4 weekly copies, one on another machine
RPO and RTO	RPO 24 hours (a day of tickets and payments could be lost), RTO 1 hour
Roles	Roles are not in a database dump. Keep pg_dumpall --roles-only next to each backup
Sensitive content	License plates, payments and staff password hashes. The dump must be stored privately

3. Backup creation

```
pg_dump -U postgres -h localhost -Fc -d grp_parkitdb -f grp_parkitdb_activity6.dump
```

Item	Value
File	grp_parkitdb_activity6.dump
Size	18,273 bytes
SHA-256	1A79AB4D7C0334016FACEF59FAE512652856C121F80EDDBC731ECFBE1F59AF56
Archive	custom format, gzip, 58 TOC entries, 5 TABLE DATA entries, 11 ACL entries, 0 DEFAULT ACL entries (the project sets no default privileges), dumped by pg_dump 18.6
Errors	None, exit code 0

4. Restoration and verification

```
createdb -U postgres -h localhost grp_parkitdb_restore
pg_restore -U postgres -h localhost -d grp_parkitdb_restore grp_parkitdb_activity6.dump
```

Restore succeeded with exit code 0 in 0.28 seconds and no messages.

Check	Original	Restored	Match
Tables, sequences, indexes	5, 5, 9	5, 5, 9	Yes
Rows (vehicle, transaction, payment, staff_user, parking_zone)	66, 19, 12, 2, 1	same	Yes
Constraints: check, foreign key, not null, primary key, unique	2, 3, 22, 5, 4	same	Yes
Table grants (parkit_runtime_grp, postgres)	11, 35	same	Yes
md5 checksum of payment	2a58980a…c7	2a58980a…c7	Yes
md5 checksum of transaction	7a2ecbaf…b3b	7a2ecbaf…b3b	Yes
md5 checksum of staff_user	0e7767ff…236eeb9	same	Yes
Sequence positions (zone, payment, staff, transaction, vehicle)	1, 12, 2, 64, 68	same	Yes
Check constraint still enforced (payment amount -1)	rejected	rejected	Yes
Application started against the restored copy as parkit_app	n/a	Login, dashboard and payment history worked. The monitor showed 7 active vehicles, the same as the restored data	Yes

5. Access control after restore

Item	Original	Restored copy
Table privileges for parkit_runtime_grp	11	11 (stored in the dump)
Database-level ACL (CONNECT for the group)	parkit_runtime_grp=c	empty (not restored)
parkit_app logs in and reads vehicle and staff_user	yes	yes (66 vehicles, 2 users)
parkit_app cannot delete payments	yes	yes (permission denied)
Object owner	postgres	postgres

The roles themselves live in the cluster, so on this server they were still present and the grants lined up. On a new server the roles would have to be created first, or pg_restore would print errors for the grants. The database-level CONNECT grant did not come back; PUBLIC still has CONNECT by default, so parkit_app could log in, but the restored database does not carry the access plan. After a restore on a new server, run the role SQL again (including the sequence grant).

6. Recovery simulation

Step	Action	Result
Before	Checksum of payment in the restored copy	12 rows, md5 2a58980a...
Incident	UPDATE payment SET amount = 0.01, then DELETE of the Cash payments, directly in the database	UPDATE 12, DELETE 8. 4 rows left, md5 9e0f29f6..., total collected 0.04
Recovery	createdb grp_parkitdb_recovery and pg_restore of the same dump	Success, exit code 0
Compare	Checksum in the recovery database	12 rows, md5 2a58980a..., the same as before the incident; total collected 720.00
Original untouched	Checksum in grp_parkitdb	12 rows, same md5

Outcome: recovered. A full restore into a new database leaves the damaged copy for comparison. Anything entered after the last backup would be lost (the RPO). Payments were chosen because they are the money records, and the app's own account cannot change them (Activity 4), so the realistic way to lose them is direct database access.

7. Evaluation and recommendations

- The project has no backup or restore instructions. Add the dump and restore commands used here to the README.
- Add pg_dumpall --roles-only to the procedure, and re-run the role SQL after a restore on a new server because the database-level grant does not come back.
- A daily dump is enough for a small lot, but a day of tickets and payments is what would be lost. If that is too much, use WAL archiving for point-in-time recovery.
- Store the dump on another machine and test a restore regularly.
- Keep config\database.php out of the backup set in Git, and store the real password separately.

8. Limitations and errors

- A logical dump captures one moment only.
- The restore ran as postgres, so objects belong to postgres, the same as the original.
- The restore gave no errors or warnings.
- The test databases grp_parkitdb_restore and grp_parkitdb_recovery are dropped after the evidence is captured.

9. Feedback for the owners

Backup and restore worked first time. Suggested additions: a backup section in the README, a roles-only dump, and re-running the role SQL after restoring on another server.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own computer, and every result in this report comes from the saved output files.